SE LAB SUBMISSION (PES2UG24CS484)

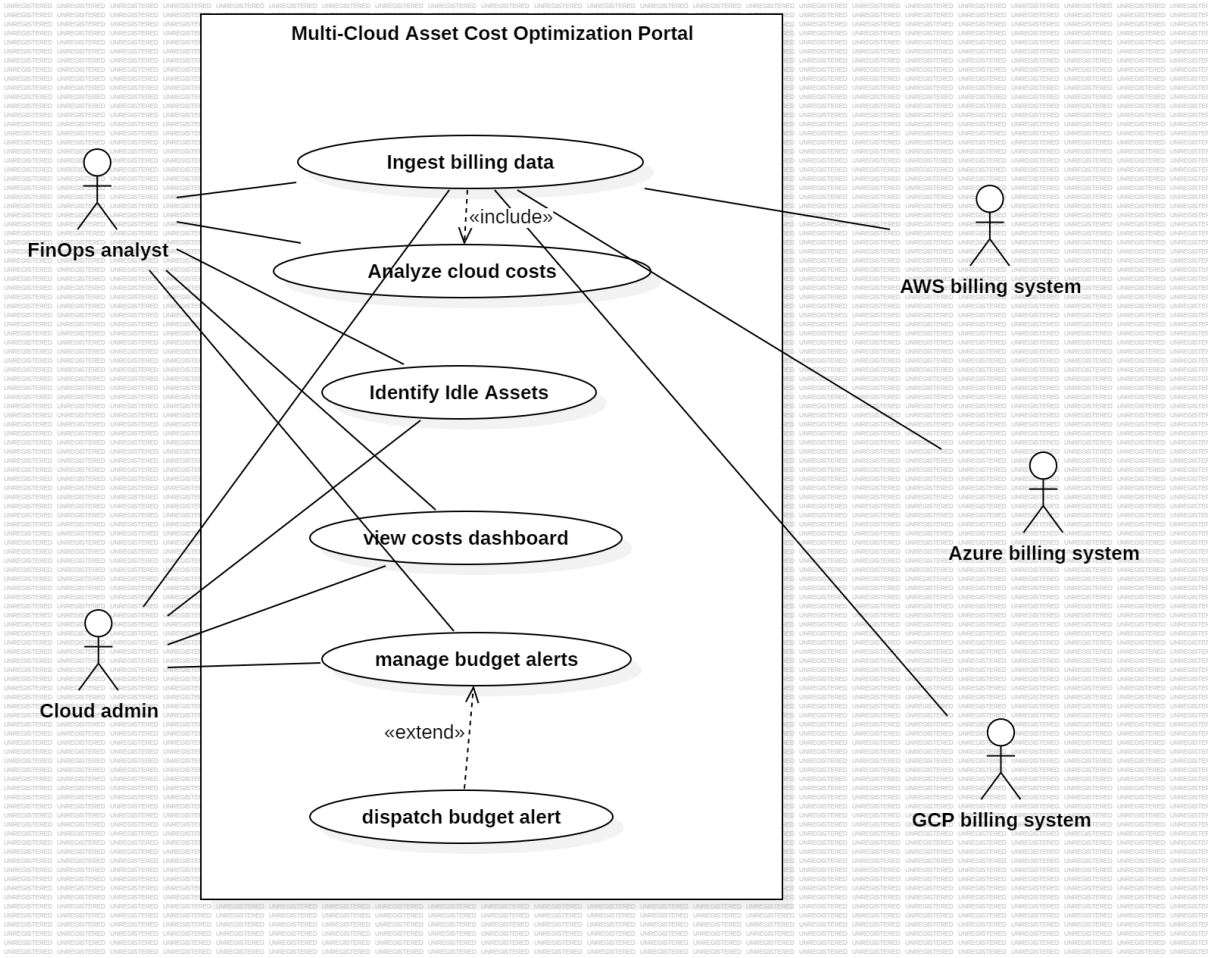
Functional and Non-Functional Requirements:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system ingests AWS, Azure, and GCP billing CSV exports, categorize costs by service and project tag, and highlight compute instances with less than 5% CPU utilization over 14 days.	High	When billing data is categorized and qualifying instances are highlighted	Cost analysis	Billing in .csv file
FR-002	Functional	The system allows the FinOps Analyst or Cloud Admin to ingest billing CSV files from AWS, Azure, and GCP.	High	A valid billing CSV from each supported provider is successfully accepted and processed	Enables data input	Specify max file size and supported CSV structure
FR-003	Functional	The system shall identify unattached storage volumes and idle compute instances from the ingested cloud data.	High	All resources meeting the defined unattached storage or idle compute conditions are displayed as optimization candidates	Finds wasted resources	Specify the criteria for an idle compute instance.

FR-004	Functional	The system shall display cloud costs categorized by cloud provider, service, and project tag through a cost dashboard.		The dashboard displays costs grouped by provider, service, and project tag.	Provides cost viability	Specify needed dashboards and charts (if needed)
FR-005	Functional	The system shall allow the FinOps Analyst or Cloud Admin to configure budget thresholds and manage budget breach alerts.	High	A user can configure a budget threshold and the saved threshold is displayed as active.	Controls spending	Specify whether thresholds apply per provider, service or project
NFR-001	Non functional (performance and security)	The portal shall parse multi-gigabyte cloud billing files and update cost dashboards in under 2 minutes while maintaining the required security standards under simulated peak load.	high	Benchmarking confirms billing files are processed and dashboards updated in less than 2 minutes under simulated peak load, while security tests pass.	Ensure speed and security	Specify file size and security standards for testing
NFR-002	Non functional (security)	The system shall restrict billing data and budget management functions to authenticated and	high	Authorized users can access permitted functions and unauthorized users are denied access	Protects sensitive data	Define permissions for each user

		authorized users				

Use case diagram:



Use case Flow:

Goal	To allow the FinOps Analyst or Cloud Admin to upload and process AWS, Azure, or GCP billing data for cost analysis.
Actor	Primary: FinOps Analyst, Cloud Admin Supporting: AWS Billing System, Azure Billing System, GCP Billing System
Main Flow	1. User selects Ingest Billing Data. 2. System displays the supported cloud providers: AWS, Azure, and GCP. 3. User selects a cloud provider and

	uploads the corresponding billing CSV. 4. System receives and validates the uploaded CSV. 5. System processes the billing data. 6. System categorizes costs by service and project tag. 7. System identifies relevant compute and storage information for further analysis. 8. System makes the processed data available to Analyze Cloud Costs. 9. System displays a successful ingestion confirmation. 10. Use case ends successfully.
Alternate Flow	Invalid Billing CSV: System detects that the uploaded CSV is invalid or unsupported. 1. System displays an error message. 2. System rejects the file and does not process it. 3. System prompts the user to upload a valid billing CSV. 4. User uploads a corrected file. 5. System validates the new file and resumes the main flow from Step 5.